

ENZO AGUIRRE

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EDUCATION

Bachelor of Science in Mechanical Engineering, Purdue University

May 2026

Bilingual IB Diploma, Markham College

Dec 2021

TECHNICAL SKILLS

CAD/CAM: Creo Parametric, SolidWorks, Siemens NX, Fusion 360, CNC, 3D prototyping and woodworking, GD&T.

ANALYSIS TOOLS: MATLAB, FEA, Excel, Python.

SPECIALTIES: Mechanical Design, Product Development, Tolerance Awareness, Testing, Fabrication.

ENGINEERING EXPERIENCE

Junior Supervisor Intern

May 2023 – Jul 2023

AAC Mining – Jaipur, India

- Worked across maintenance, development, and supply chain operations in an active mining environment
- Coordinated with Caterpillar, Epiroc, and Sandvik support teams on equipment maintenance and readiness
- Contributed to maintenance coordination efforts that helped reduce machine downtime by 20%
- Evaluated equipment availability, site logistics, and operational feasibility for multiple prospective mining sites

Powertrain Designer

2023 - 2025

Purdue University - ASME EV Race Kart Team – West Lafayette, IN

- Designed drivetrain and supporting mechanical components for EV and ICE kart platforms using CAD
- Applied load, torque, packaging, and manufacturability considerations to improve durability and serviceability
- Supported fabrication, assembly, and iterative testing during vehicle build cycles
- Collaborated with cross-functional mechanical and electrical teams on integration, fitment, and reliability

PROJECT EXPERIENCE

Performance Brake Assembly Design

- Designed a sport-sedan brake assembly including 340 mm rotor, fixed multi-piston caliper, pad geometry, piston layout, and mounting features.
- Performed FEA for clamp loading, deformation, and stress concentration analysis
- Iterated geometry toward improved stiffness, reduced mass, and manufacturability
- Applied packaging and mounting constraints during design iterations

Rotary Engine Prototype

- Designed and fabricated a scaled rotary engine prototype using CAD, 3D printing, CNC machining, and sensor-based testing.
- Used CNC machining, 3D printing, and CAD modeling to manufacture components
- Integrated sensors and MATLAB analysis for efficiency testing

Custom 7-Driver Speaker

- Designed and built custom 3-way speaker system with seven drivers, crossover layout, and tuned enclosure
- Integrated amplification, driver packaging, and cabinet construction for improved acoustic performance
- Balanced sound quality, manufacturability, packaging, and structural design constraints

Capstone Project: Adaptive Elbow Rehabilitation (AER)

- Led lifecycle from concept to prototype, guiding 5 engineers through 300+ hours of electromechanical design
- Developed and programmed a brushless DC powered reel using CAN bus systems to provide variable slow speed resistance
- Designed brace structure using Creo Parametric and used FEA analysis for stress concentration and torque load